

Historical Scientific Supplement

Neural Differential Utility: Accepted Dynamic Service Contracts and Certified Value-Gradient Learning

Preserved material accompanying Revision R12, September 22, 2026

This archival supplement preserves, without scientific deletion, the earlier continuous-time, diffusion, and computational derivations formerly embedded in the electronic companion. It is separately labeled so that historical evidence is not mistaken for verification of the current accepted nonlinear tree model. The complete source and all cited predecessors remain in the revision package.

H.1 Comparator geometry and inner optimization

H.1.1 What an exposed-face argument proves

Fix a state and a physical action, and let the continuation-value vector range over an open subset $U \subset \mathbb{R}^d$. A finite family of admissible valuation controls induces

$$F(v) = \max_{j=1,\dots,J} \{g_j + \beta p_j^\top v\}. \quad (\text{H.1})$$

Here p_j is a transition probability vector and g_j includes the control-dependent payoff and adjustment terms. The restrictions on the comparator must be specified before any non-equivalence statement is made.

Proposition H.1.1 (Two exposed slopes). *Suppose $\beta > 0$ and there are open subsets $U_1, U_2 \subset U$ on which distinct indices j_1, j_2 are the unique maximizers, with $p_{j_1} \neq p_{j_2}$. No single affine primitive $g + \beta p^\top v$ equals F throughout U . Nevertheless, an ordinary Markov decision problem with the valuation index admitted as an action represents F exactly.*

Proof. On U_i , the derivative of F is βp_{j_i} . Equality with a single affine function on an open set would require its derivative to equal both distinct vectors. This is impossible. The last statement follows by taking the ordinary action set to be the original physical action paired with j . $\square$

Different intercepts alone do not establish the premise: if all slopes coincide, the largest intercept gives a single affine envelope. Nor does the proposition say that a continuum of affine functions must have a nonsmooth envelope. For example, with strictly positive reference probabilities p_i and $\rho > 0$, set

$$Z = \sum_i p_i e^{\rho \beta v_i}, \quad q_i^* = p_i e^{\rho \beta v_i} / Z.$$

For any probability vector q ,

$$\beta q^\top v - \rho^{-1} \text{KL}(q||p) = \rho^{-1} \log Z - \rho^{-1} \text{KL}(q||q^*). \quad (\text{H.2})$$

Nonnegativity of relative entropy proves the Gibbs variational identity. Entropic continuation is therefore an exact member of a sufficiently rich payoff–kernel envelope family. A fixed entropic risk parameter does not exclude this representation. The sign of the risk parameter matters: the displayed maximization convention is for $\rho > 0$; the corresponding negative-parameter convention reverses the variational optimization. No general exclusion of risk-sensitive or robust control is claimed.

Exogenous preference processes and fixed multiplier formulations can be compared only after fixing their admissible transition/payoff families. A restricted family may fail to contain a particular joint policy; that is not a theorem that every conventional stochastic-control formulation fails. Expanded-action equality in Proposition EC.4.1 of the main paper is the definitive representation check.

H.1.2 Conditional strongly concave reductions

A separate sufficient computational condition, useful in continuous-control models, is strong concavity of a particular inner Hamiltonian. It is not needed for the exact envelope algorithm or the quadratic contract-grid bound.

Proposition H.1.2 (Inner-selector residual bound). *Let f be differentiable and μ -strongly concave on a compact convex set $\Theta \subset \mathbb{R}^d$. Let $\widehat{\theta} \in \Theta$ satisfy*

$$\nabla f(\widehat{\theta}) = n + e, \quad n \in N_{\Theta}(\widehat{\theta}), \quad \|e\| \leq \varepsilon,$$

where $N_{\Theta}(x) = \{n : n^{\top}(y - x) \leq 0 \ \forall y \in \Theta\}$. Then the unique maximizer θ^* satisfies

$$\|\theta^* - \widehat{\theta}\| \leq \varepsilon/\mu, \quad f(\theta^*) - f(\widehat{\theta}) \leq \varepsilon^2/(2\mu).$$

For $f(\theta) = b^{\top}\theta - \theta^{\top}R\theta/2$ with $R \succ 0$, the optimizer is the projection of $R^{-1}b$ onto Θ in the R norm. In the scalar interval case it is ordinary clipping.

Proof. Strong concavity and the normal-cone inequality give

$$f(\theta^*) - f(\widehat{\theta}) \leq \varepsilon d - \mu d^2/2 \leq \varepsilon^2/(2\mu), \quad d = \|\theta^* - \widehat{\theta}\|.$$

For the distance bound, combine the strong monotonicity of $-\nabla f$ with the first-order optimality condition at θ^* and the stated approximate normal-cone condition. They imply $\mu d^2 \leq \varepsilon d$. Completing the square in the quadratic objective proves the projection statement. $\square$

Thus a physical-action approximation with gap δ_a , combined with an inner residual ε , has total maximization gap at most $\delta_a + \varepsilon^2/(2\mu)$ when these hypotheses hold uniformly. Positive maintenance cost by itself does not establish global concavity after taking the stock envelope and adding continuation. The main paper avoids that unsupported inference.

H.2 Physical metrics and the historical newsvendor argument

H.2.1 Restricted critical-fractile mismatch

Consider the separate physical cost

$$C_z(S) = h\mathbb{E}(S - D_z)^+ + p_z\mathbb{E}(D_z - S)^+, \quad h, p_z > 0.$$

Assume the relevant stock interval has a density $f_z \geq \underline{f}_z > 0$ and the interior minimizer S_z^* satisfies $F_z(S_z^*) = p_z/(h + p_z)$. A fixed service parameter uses a common effective shortage coefficient $\bar{p}$ and selects $\bar{S}_z$ at fractile $\bar{p}/(h + \bar{p})$. Then

$$C_z(\bar{S}_z) - C_z(S_z^*) \geq \frac{(h + p_z)\underline{f}_z}{2}(\bar{S}_z - S_z^*)^2. \quad (\text{H.3})$$

Indeed $C'_z(S_z^*) = 0$ and $C''_z = (h + p_z)f_z$ on the connecting interval; integrate the second derivative twice. A positive-weight collection of regimes yields a strict expected gap whenever at least one selected stock differs from its physical optimum. Bounds on density and on quantile separation make this a stable mismatch statement.

This argument is useful and is retained. Its comparator is a common critical-fractile parameter, not an unrestricted state-aware inventory controller. Selecting a state-dependent parameter afterward so that its penalty equals p_z does not prove that an endogenous net-reward optimizer selects it. The actual-optimal-policy accounting theorem in Section EC.2 of the main paper supplies that missing distinction. In the recorded relaxation the joint policy's physical cost is computed from its own exact optimizing actions, not from a physical-cost-matching contract inserted by hand.

H.2.2 The actual optimized ledger

Write $W^\pi = K^\pi - C^\pi$, with all entries discounted under the same law. The exact results give

$$W^J - W^F = 0.944747060878850\dots, \quad C^F - C^J = 0.213203951173630\dots$$

Their difference is the increase in net contract transfers. The physical-cost oracle attains approximately 8.506094529; the joint policy costs 9.374173780. Thus improved net reward does not imply physical optimality. The oracle's contract-one implementation is fixed by the recorded tie convention, and its net cash flow is not the objective it minimizes.

Fill rate is discounted filled demand divided by discounted demand, not a reward rescaling. It declines from 0.9259259259... under the frozen contract to 0.8821088614... under the joint contract. This decline is part of the result, not a failed run to discard. The full output also retains undiscounted filled, lost, and total demand. Historical columns named `avg_cost` that include adjustment or valuation-dependent terms remain total-cost diagnostics; they are not relabeled as physical cost.

H.3 Continuous-time derivations and boundary controls

H.3.1 Value volatility and the adjoint

For shared Brownian noise, the general relation is

$$Z = \sigma_c^\top P_c + \sigma_h^\top P_u,$$

not two independently invertible observations. When $\sigma_h = 0$, the internal gradient is absent from this volatility. If the external block also lacks full row rank, some components of P_c are unobserved as well. If the physical action affects internal drift, the term $P_u^\top \partial_a h$ enters the physical first-order condition; the degeneracy need not be confined to the contract action.

For completeness, suppose the value is sufficiently smooth, diffusion is control-independent, the optimal selector satisfies the Hamiltonian envelope condition, and all required derivatives and stochastic integrals exist. Before exit, set $P = \nabla V$ and $Q = D^2 V \Sigma$. Differentiating the HJB equation and applying Itô's formula gives

$$dP_s = - \left[b_x^\top P_s + r_x + \sum_j (\sigma_j)_x^\top Q_{s,j} - \rho P_s \right] ds + Q_s dB_s. \quad (\text{H.4})$$

Here derivatives of the primitives hold the control fixed; the envelope condition removes derivatives of the maximizing selector. Along paths reaching the terminal face without prior exit, $P_T = \nabla G$. A Dirichlet exit payoff fixes the boundary value and tangential derivatives when smooth, but does not fix the normal derivative. Consequently one must not impose $P_\tau = \nabla \psi$ in all directions merely from the Dirichlet value condition. The stopped value-process equation, rather than such an extra adjoint terminal condition, is used in the diffusion certificate below.

Adding an independent diffusion $\sqrt{2\varepsilon}I$ to the internal block gives $Z_u^\varepsilon = \sqrt{2\varepsilon}P_u$. Its inverse has squared amplification $1/(2\varepsilon)$ under additive volatility error. A direct gradient avoids this inverse but must still solve the value or adjoint problem. The retained least-squares parity result tests naturally generated data rather than adding a fixed error floor to the volatility target.

H.3.2 Why both boundary faces and both comparisons are necessary

A terminal condition alone does not determine a compact spatial problem. On $x \in [1, 2]$, let

$$w(t, x) = (1 - t)^{-1/2} \exp\{-x^2/[4(1 - t)]\}, \quad t < 1, \quad w(1, x) = 0.$$

It satisfies $w_t + w_{xx} = 0$ and extends smoothly to zero at the terminal face on this spatial interval, yet it is nonzero in the interior. The zero function has the same terminal data and interior equation. Their distinct lateral values explain the difference. An interior and terminal residual cannot control that difference without boundary information.

A separate one-step example shows why a policy bound needs both critic comparisons. There are two actions leading deterministically to distinct terminal states, with true terminal reward zero. Action rewards

are 0 and 1.9. A terminal critic equal to +1 at the first state and -1 at the second has terminal sup error one. Its greedy action selects the first action, yielding true policy loss 1.9. The preceding critic can be its exact Bellman update, so the Bellman residual and greedy gap relative to that critic are zero. A policy bound of only one terminal error is false; two terminal errors suffice. This example is executed as an exact rational regression.

The diffusion theorem separately bounds value error and the critic-to-actor value error. For an actor-induced residual, the difference between the actor and optimal Hamiltonians must also be charged. A Neumann defect, reflected boundary local time, and a state-constraint viscosity inequality are not interchangeable with the Dirichlet mismatch used here.

H.3.3 From finite validation to a uniform bound

Suppose an independently established Lipschitz constant L_R bounds the interior residual over the specified compact domain, and validation nodes form a cover of fill distance h_R . Then

$$\sup_D |R| \leq \max_{x_i} |R(x_i)| + L_R h_R.$$

Terminal and lateral defects require their own covers and constants. An actor gap likewise requires a certified action cover and Hamiltonian continuity modulus. Training loss, a sampled percentile, or a ratio against another network supplies none of these constants by itself. A differentiable network does not automatically supply a useful global residual bound: derivative growth and domain coverage must be bounded.

A vanishing-viscosity value conclusion can be made under explicit comparison, common boundary attainment, uniform boundedness and equicontinuity, and locally uniform convergence of the HJB operators. Compactness gives subsequential value limits; stability of the sub- and supersolution inequalities identifies a solution of the limiting boundary problem; comparison identifies all such limits. Uniform value convergence alone does not prove convergence of gradients or recovery of the missing shadow price. These conditional observations preserve the original regularization motivation without using it as a substitute for the absolute certificate.

H.4 Continuous contract reconfiguration and learned value gradients

The same operating primitives yield a continuous-review model in which contract persistence affects a genuine control problem. Let the demand regime be a continuous-time Markov chain with generator Q , and let the installed tier evolve as $\dot{q}_s = v_s$, $|v_s| \leq \bar{v}$. At $q = 0$ only $v \geq 0$ is admissible; at $q = 1$ only $v \leq 0$ is admissible. There is no exit payoff or reflected diffusion in this model. The operating reward rate is

$$R_z(q) = r_z q - m q^2 + U_z(q), \quad \mathcal{W}(t, z, q) = \sup_v \mathbb{E} \int_t^T e^{-\rho(s-t)} \{R_{z_s}(q_s) - \lambda v_s^2\} ds.$$

Stock is selected by (EC.14) at the current installed tier. Finite-stock demand primitives make R a piecewise-quadratic, generally nonsmooth function. The value is unknown; it is not chosen and then used to manufacture

the reward. An optional customer commitment can be dualized by replacing R with the similarly reduced reward $g + \mu f + \eta u - \chi C$. The numerical learning experiment uses the provider relaxation, separately from the exact accepted-contract comparison above.

Proposition H.4.1 (A locally consistent continuous-review embedding). *Consider review intervals Δ , allowed changes $\theta = q + v\Delta$, transition $I + Q\Delta$, discount $e^{-\rho\Delta}$, operating reward $\Delta[r_z\theta - C_z(S) - \kappa\theta L_z(S) - m\theta^2]$, and adjustment cost $\lambda(\theta - q)^2/\Delta$. For a bounded test function smooth in time and q , the one-step dynamic programming operator, after subtracting the test function and dividing by Δ , converges to*

$$w_t + R_z(q) + (Qw)_z - \rho w + \sup_{v \in A(q)} \{vw_q - \lambda v^2\}, \quad (\text{H.5})$$

where $A(q)$ is the inward velocity interval. For tests with bounded required second derivatives, the local consistency error is $O(\Delta)$.

Proof. On each review, $\lambda(\theta - q)^2/\Delta = \lambda v^2\Delta$. Taylor expand $w(t + \Delta, z', q + v\Delta)$ and $e^{-\rho\Delta}$, and use $I + Q\Delta$. Products of regime changes and q changes are of order Δ^2 . The operating reward at $q + v\Delta$ differs from that at q by $O(\Delta)$ before multiplication by Δ , since the compact-menu reward is uniformly Lipschitz in its tier. The remainder is uniform over the compact velocity set, so taking the maximum preserves convergence. $\square$

This is an explicit flow-rate embedding with a rescaled adjustment coefficient and bounded revision speed. It is not a claim that sending the period length to zero in the unscaled eight-period model leaves its economic units unchanged. The connection is through the same premium, demand, liability, stock, and maintenance primitives.

In the interior, the optimal instantaneous velocity for a smooth value is

$$v^*(t, z, q) = \text{proj}_{[-\bar{v}, \bar{v}]} \frac{\mathcal{W}_q(t, z, q)}{2\lambda}. \quad (\text{H.6})$$

At the endpoints the projection uses $A(q)$. This makes a learned derivative operational: a small value error need not give a small revision-speed error. A scalar critic w_φ supplies the compatible derivative $\partial_q w_\varphi$ rather than an independently fitted vector field.

H.4.1 Verification and discrete certification

Theorem H.4.2 (Verification for the confined switching model). *Assume bounded rewards, a finite-state generator, compact velocities, well-posed admissible confined trajectories, and measurable arbitrarily accurate Hamiltonian selectors. Let each w_z be C^1 in (t, q) with bounded derivatives. Suppose the absolute residual in (H.5) is at most δ_H everywhere, terminal error is at most δ_T , and an admissible actor loses at most δ_A in the velocity Hamiltonian. Then*

$$|w - \mathcal{W}| \leq \delta_T + H_\rho \delta_H, \quad 0 \leq \mathcal{W} - \mathcal{W}^A \leq 2\delta_T + H_\rho(2\delta_H + \delta_A),$$

where $H_\rho = \int_t^T e^{-\rho(s-t)} ds$.

Proof. Apply the discounted chain rule with the compensated regime-jump martingale to $w(s, z_s, q_s)$. Its drift is $w_t + v w_q + Q w - \rho w$. Boundedness gives integrability. The upper residual inequality bounds every policy; an arbitrarily accurate maximizing selector gives the reverse value inequality. For the specified actor charge its Hamiltonian gap as well. Add the two critic comparisons for the policy bound. The process remains in $[0, 1]$ and has no boundary local-time or stopping payoff; the inward restriction is part of the maximization, including at endpoints. $\square$

The stopped Dirichlet theorem, controlled-diffusion Hessian term, lateral-boundary defects, and singular-volatility examples from the earlier development are retained in the Electronic Companion (EC). They concern a different boundary mechanism. The present theorem does not remove lateral errors from a stopped problem.

For computation use $q_i = ih$ and $\Delta = T/N_t$ satisfying $\Delta(\bar{v}/h + \max_z(-Q_{zz})) \leq 1$. A positive speed moves to $i + 1$ with probability $\Delta v/h$, a negative speed to $i - 1$ with probability $-\Delta v/h$, and a regime jump to $z' \neq z$ with probability $\Delta Q_{zz'}$. Remaining probability stays put. At endpoints impose inward speeds. With $d = e^{-\rho\Delta}$, its exact Bellman operator is

$$(\mathcal{T}_h W)_{z,i} = \Delta R_z(q_i) + d\{W_{z,i} + \Delta(QW)_{z,i}\} + \Delta \max_{v \in A(q_i)} \{d[v^+ D^+ W + v^- D^- W] - \lambda v^2\}, \quad (\text{H.7})$$

where $v^+ = \max(v, 0)$ and $v^- = \min(v, 0)$. Optimize each sign branch as a scalar quadratic and compare them. These are exact continuous-speed maximizations for a specified finite-state Markov chain. For a smooth test function the upwind spatial truncation is generally $O(h)$ and the temporal truncation $O(\Delta)$; neither vanishes merely because the original manufactured test was quadratic and time independent. The grid refinement below is an observed convergence study, not a proved continuum error rate for this nonsmooth value.

Proposition H.4.3 (A complete finite-chain learned-policy budget). *Let $w_{N_t} = 0$, $e_k = \|w_k - \mathcal{T}_h w_{k+1}\|_\infty$, and $a_k = \|\mathcal{T}_h w_{k+1} - \mathcal{T}_h^{A_k} w_{k+1}\|_\infty$. The actor's initial loss against the exact chain optimum is at most*

$$\sum_{k=0}^{N_t-1} d^k (2e_k + a_k). \quad (\text{H.8})$$

Proof. Both Bellman and fixed-actor operators are monotone d -contractions in sup norm because their continuation coefficients are transition probabilities. Backward induction gives a critic-to-optimum error of at most $\sum d^k e_k$ and a critic-to-actor error of at most $\sum d^k (e_k + a_k)$. Add them. $\square$

Every maximum in this certificate is over the complete finite chain, not a sampled loss. The exact formula is a chain statement; its floating-point evaluation is reported as a numerical budget, not an interval-arithmetic enclosure. Applying Theorem H.4.2 to the continuum would additionally require uniform residual bounds between validation points. A refinement difference does not provide those bounds.

H.4.2 A genuine training problem

We train the scalar architecture

$$w_\varphi(t, z, q) = 8(1 - t/8) \text{MLP}_\varphi(t/8, q, \text{onehot}(z)),$$

with widths 5–32–32–1 and two hyperbolic-tangent layers, totaling 1,281 trainable parameters. The terminal value is exactly zero by construction. The teacher is the nonmanufactured inventory-contract problem on a 160-interval, 803-step chain. Values and centered-difference derivative labels are interpolated at 2,048 independent time–tier–regime draws for each of seeds 17 and 29. Both methods use the same draws, initial weights, minibatch sequence, and 6,000 Adam steps. Learning rates are 0.003, multiplied by 0.35 at steps 3,000 and 5,000. The objectives are

$$\mathcal{L}_0 = \frac{1}{N} \sum_j (w_\varphi(x_j) - Y_j)^2, \quad \mathcal{L}_1 = \mathcal{L}_0 + \frac{0.1}{N} \sum_j (\partial_q w_\varphi(x_j) - G_j)^2.$$

Derivative-informed training is an established method (Czarnecki et al., 2017); here its role is to train the derivative entering the contractual revision decision. The second loss uses additional derivative information, not an equivalent reparameterization of identical targets. The gradient labels are numerical teacher estimates, not exact continuum derivatives.

Deployment uses the projected derivative actor, with the one-step discount factor appropriate to (H.7). We evaluate on a separate 80-interval, 403-step chain and compute both its Bellman reference and the entire actor value recursion. In addition to realized chain losses, we evaluate (H.8) at all 98,172 time–state points, reporting its deliberately distinct worst-case budget. This is actual optimization of randomly initialized networks, rather than a hinge compilation. The constructive cubic-ReLU result remains useful as a different, deterministic approximation theorem. The learning experiment establishes an operational use of compatible gradients on these instances, not superiority over exact dynamic programming or a new general neural convergence theorem.

H.5 Retained R6 computational study

We retain the recorded canonical instance: $T = 8$, $\beta = 0.97$, $q_0 = 0.5$, $z_0 = 1$, and $D_z = z + D_0$, with $D_0 = 0, 1, 2, 3$ having probabilities 0.1, 0.3, 0.4, 0.2. The transition matrix is

$$P = \begin{pmatrix} .70 & .25 & .05 \\ .15 & .70 & .15 \\ .05 & .25 & .70 \end{pmatrix}.$$

Stock is $0, 0.5, \dots, 6$ and the primary tier grid is $0, 0.1, \dots, 1$. The cost coefficients are $c = 0.3$, $h = 0.6$, $p = 0.65$, $\kappa = 1.2$, $m = 1.5$, and $\lambda = 0.6$. Premiums satisfy $r_z = 0.6 + 1.4z$. Thus the compatibility sufficient bound is $1.4 \geq 1.2$. There are 264 nonterminal states and 37,752 full state–action pairs. The customer extension fixes $\nu = 2$ and a common retainer of 6; these do not change any historical incremental provider

payoff. Every service measure in the new comparisons is a ratio of expected discounted totals, not an average of random pathwise ratios.

H.5.1 Optimized baselines and marginal information value

Table 1 separates better static tier selection from dynamic information. The continuous static optimum improves on the previously reported 0.5 contract by 0.076946642417. The 0.1-grid dynamic policy then exceeds this optimized continuous static value by 0.867800418462. The corresponding improvement over the best static grid tier 0.6 is 0.871452959339. A time-only schedule adds zero on this symmetric instance; Proposition EC.1.1 gives the same value for inherited-only rules.

The optimized time–regime rule has value -5.052903963837 , yielding 0.820859550086 above the best static grid contract. Adding the inherited tier to that information set contributes a further 0.050593409253. The regime-only optimization uses binary shared-action constraints, not an oracle allowed to condition secretly on q . Its mixed-integer solve reports matching primal and dual values to numerical precision, with a zero reported relative gap. This solver certificate is numerical; the full-policy and continuous-static calculations have separate exact rational replays. Unreachable-state choices do not affect any reported initial value.

Table 1: Optimized contract classes and successive information increments

Contract rule	Reward	Increment
Frozen 0.5	-5.947058	–
Best continuous static	-5.870111	0.076947
Best grid static / time / inherited-only	-5.873764	–
Time and regime	-5.052904	0.820860
Full (t, z, q)	-5.002311	0.050593

The first increment improves the frozen tier; the last two follow the nested grid-class chain. Continuous and grid static values are reported separately. Stock uses observed regime in every row.

H.5.2 A service and participation frontier

Table 2 reports the full-stock occupation program. Requiring the static fill level raises the dynamic cost to the same physical cost as the static policy. Requiring the static customer’s utility as well reduces provider reward further, but a gain of 0.625140311972 remains against the *best continuous* static tier. Adding the static physical-cost cap leaves this solution unchanged. Total provider profit after the common retainer increases from 0.129889026953 to 0.755029338925; customer surplus after the same retainer is unchanged. The exact certificate in Section EC.1.2 verifies all three equalities, not just the objective difference.

More stringent service is expensive: at a 0.95 fill floor with the same customer reservation utility, provider incremental reward is -6.266872787932 , below the static reference, and physical cost rises. This point is feasible for the operating commitment program but fails a zero provider outside option with retainer 6. Its purpose is to show the service-price frontier, not to recommend an unaccepted agreement. Thus enlarging dynamic flexibility does not make arbitrarily demanding service commitments free.

Table 2: Provider reward, physical cost, customer utility, and discounted fill

Protocol / requirement	W	C	U	Fill
Continuous static reference	-5.870111	9.587378	28.657506	0.925926
Provider relaxation	-5.002311	9.374174	25.449343	0.882109
Fill 0.90	-5.021190	9.461228	26.049111	0.900000
Fill 0.92	-5.080921	9.558544	26.756726	0.920000
Static fill	-5.101718	9.587378	26.967723	0.925926
Static fill and utility	-5.244971	9.587378	28.657506	0.925926
Static fill, utility, cost	-5.244971	9.587378	28.657506	0.925926
Fill 0.95 and utility	-6.266873	11.110185	28.657506	0.950000
Fill 0.98 and utility	-7.688163	13.007837	28.657506	0.980000

The last column is expected discounted filled demand divided by expected discounted demand. Incremental utilities exclude a common retainer. At a 0.95 or 0.98 floor the displayed operating solutions need a larger retainer to satisfy a zero provider outside option.

H.5.3 Sensitivity and computational scaling

The parameter study separately varies λ, κ, m , the premium slope, discounting, and regime persistence, leaving all other primitives fixed. Table 3 displays the adjustment-cost slice. Discounted squared movements decline from 0.584931 at $\lambda = 0$ to 0.051710 at $\lambda = 4.8$, consistently with Proposition H.10.2. The adaptive gain also decreases in this particular slice, but that second ordering is not asserted by the proposition. All recomputed static comparators optimize their own tier. The full CSV records statewise monotonicity and outcomes for every parameter slice, including cases in which the sufficient compatibility inequality fails but the selected policy happens to remain monotone.

Table 3: Adjustment cost, adaptive value, and squared movement

λ	Full reward	Best static grid	Gain	Movement
0.000000	-4.714286	-5.867764	1.153478	0.584931
0.150000	-4.800467	-5.869264	1.068797	0.541675
0.300000	-4.880586	-5.870764	0.990178	0.520250
0.600000	-5.002311	-5.873764	0.871453	0.346310
1.200000	-5.162935	-5.879764	0.716828	0.192041
2.400000	-5.339598	-5.891764	0.552166	0.102878
4.800000	-5.515444	-5.915764	0.400320	0.051710

Each static comparator is reoptimized at its own coefficient. Movement is the expected discounted sum of squared tier changes, without multiplying by the coefficient.

We also solve the quadratic model and the nonquadratic potential $\phi(q) = q^2 + 0.2q^4$ on 32, 128, 512, and 2,048 intervals. Upper-hull and exhaustive floating implementations agree within 1.8×10^{-15} at every tested value and choose identical policies. At 2,048 intervals the full eight-period calculation reduces 100,761,624 pairwise comparisons to 49,176 inserted lines, excluding stock/continuation preprocessing. Measured speedups are approximately 22 and 13 for the two potentials on this CPU. At small grids vectorized enumeration is faster than the Python hull; the timing claim is not universal. The linear algorithmic operation count and the measured implementation costs are reported separately.

H.5.4 An operational continuous problem and actual neural training

For the connected continuous problem use $T = 8$, $Q = P - I$, $\rho = -\log(0.97)$, $\bar{v} = 0.5$, and the same inventory and tier-reward coefficients. Refining from 20 to 160 tier intervals changes the initial chain value from -5.193740808 to -5.158941235 ; the intermediate values are -5.173537813 and -5.163828702 . Both the mesh and the time step change, and the problem is nonmanufactured. These values are not a comparison with the original jump-adjustment optimum: the rate model has an explicit speed limit and different adjustment units.

Table 4 reports all four 6,000-step training runs. Adding numerical derivative targets reduces the evaluated chain-policy loss for both paired seeds, from 0.029062 to 0.014750 and from 0.020323 to 0.006948. Training costs are higher for derivative supervision; the labels contain additional information. Two seeds do not support a universal statistical superiority claim. Worst-case residual budgets remain substantially larger than observed initial losses; they are displayed rather than substituted with the smaller training errors. The complete numerical certificate scans all 98,172 evaluation points. There is no claim of a uniform continuum neural-HJB error enclosure from these finite-chain calculations.

Table 4: Actual learned-critic deployment on the operational evaluation chain

Seed	Training targets	Policy loss	Value RMSE	Residual budget
17	Value	0.029062	0.055700	5.337046
17	Value + gradient	0.014750	0.036387	2.903095
29	Value	0.020323	0.042422	4.246878
29	Value + gradient	0.006948	0.024725	2.183179

Both methods use 2,048 identical sample locations and 6,000 matched optimization steps per seed. Derivative training receives additional numerical derivative labels. The budget is the floating evaluation of the full finite-chain formula, not a continuum certificate; it is intentionally not replaced by the observed initial policy loss.

H.6 Operational implications and evidence scope

Three decisions should be separated. First choose the counterparty’s commitment and reservation utility, rather than allowing the provider to price its own success criterion. Then optimize the contingent contract against an optimized static alternative, charging the same installation cost. Finally determine the information and computational representation needed to implement that protocol. In the canonical case, regime adaptation generates most of the reward increment, inherited-tier information adds a smaller strictly positive amount, and customer-preserving adaptation remains beneficial after service degradation has been ruled out.

The mathematical identity of NDU here is explicit: externally priced joint physical and persistent-contract control, with compatible value gradients for continuous reconfiguration and independently evaluated error budgets. Its benefits are not a separation from expanded-action Markov decision processes. The structural contribution is the combination of primitive monotonicity, implementable switching thresholds, dual-coordinate envelope reduction, and an accepted-contract comparison that can be verified without relying on neural performance. The learning component supplies an actually trained derivative-based deployment rule for the connected operational model.

All earlier valid material is retained. The physical-cost oracle, payoff–kernel geometry, entropic identity, strong-concavity residual lemma, singular-volatility conditioning, stopped HJB theorem with lateral errors, manufactured-solution calculation, and historical neural diagnostics remain in the EC or their original archival paths. The identical-data least-squares result continues to report parameterization parity. These materials are not silently pooled with the new contract-learning experiment.

Reproducibility. The new entry points are `compute.py`, `exact_certificate.py`, `envelopes.py`, and `continuous_learning.py` under `revisions/or-r6-20260921/`. They generate the static candidate audit, restricted-policy comparison, full-stock service frontier, exact feasible/dual certificate, sensitivity runs, hull scaling, nonmanufactured refinement, trained weights, and exhaustive learned-policy evaluations. The earlier integer and independent Fraction implementations remain unchanged and are used to replay the canonical optimum. Exact arithmetic certificates, numerical optimization diagnostics, and finite-chain learning budgets are labeled separately. No empirical customer data or market calibration are claimed for these synthetic instances.

H.7 Execution details and constructive neural evidence

H.7.1 Frozen inputs and exact arithmetic

The experiment reads the original R2 protocol and verifies its SHA-256 before computing. The protocol bytes are unchanged. R3’s recorded conventions specify time indices, the oracle’s contract-one tie, duplicate gate labels, paired batch inference, and identical-data least squares. The contract-grid theorem, smooth neural compilation, and joint space–time–action study are additional analytic or deterministic calculations; they do not replace any recorded method or change its parameters.

For contract grid denominator n , write $S = s/2$, $\theta = j/n$, and inherited $q = i/n$. Let $w_d = (1, 3, 4, 2)$ and $D = z + d$. Define integer quantities

$$O = \sum_d w_d (s - 2D)^+, \quad L = \sum_d w_d (2D - s)^+, \quad C_{\text{num}} = 60s + 12O + 13L.$$

Expected physical cost is $C_{\text{num}}/400$. Expected net stage reward, on denominator $400n^2$, has numerator

$$80(3 + 7z)jn - C_{\text{num}}n^2 - 24jLn - 600j^2 - 240(j - i)^2. \quad (\text{H.9})$$

The transition numerator matrix on denominator 20 is

$$\begin{pmatrix} 14 & 5 & 1 \\ 3 & 14 & 3 \\ 1 & 5 & 14 \end{pmatrix}, \quad \beta = 97/100.$$

With r periods remaining, values use common denominator $400n^2 2000^{r-1}$. Thus all action comparisons use integers with no numerical tie tolerance. A separately written implementation reconstructs expected

costs from the demand realizations using `Fraction` arithmetic and enumerates the actions independently. It checks all 264 state–time values and policies for each of the five methods. It also verifies that the tie-selected pair is componentwise greatest among all maximizers.

The primary expanded-action comparator uses a flattened list of 143 physical-contract pairs. Its equality with the joint solver is exact at every state, not merely agreement after rounding. The dummy gate enumerates 11 duplicate labels per stock choice, but the economic contract and next inherited state stay at one half. It therefore has 13 distinct economic choices, not 143. Its output equals the frozen-contract solution exactly.

The 416 neighboring-state monotonicity checks support the displayed instance; the general result is the proof in the main paper. The minimum observed adjacent cross difference of the exact value is $9/250$. The exact upper hull is evaluated at all 264 primary states. The hull weights and rational switch locations are stored in `exact_envelopes.json`.

H.7.2 Simulation and uncertainty

Each of 64 independent batches contains 4,096 episodes. A NumPy PCG64 generator is seeded with $2026092300 + b$, $b = 0, \dots, 63$. Demand and regime uniforms are generated before all policy evaluations, and the same exogenous histories are used by all methods within a batch. Confidence intervals use the two-sided Student quantile with 63 degrees of freedom applied to the batch means; comparisons use paired batch differences. They quantify simulation error, not training variability or external validity. No simulated sign determines which methods are retained.

The paired net-reward increase over the frozen contract is approximately 0.943729, with 95% interval $[0.939821, 0.947636]$. The exact expectation is 0.9447470609... The paired physical-cost difference, joint minus frozen, is approximately -0.212112 , with interval $[-0.214933, -0.209291]$. The exact difference is $-0.2132039512\dots$. All batch ledgers, not only aggregate intervals, are retained.

H.7.3 Identical-data least-squares parity

The conditioning batches use seed $2026092400 + b$ and 2,048 pairs (u, ξ) . The same pairs are reused across the four noise levels within a batch. Let $X = [u^2/2, 1]$ and $X_Z = [u^2/(2\sqrt{2\varepsilon}), 1]$. For nonzero ε ,

$$X_Z = X \operatorname{diag}((2\varepsilon)^{-1/2}, 1).$$

Their fitted function spaces coincide. With full column rank, unregularized least squares gives $\hat{\zeta} = \sqrt{2\varepsilon} \hat{k}$ and the same intercept in exact arithmetic. Hence the inferred shadow-price functions coincide. Conditional expectation of the observed payoff is $u^2/2 + \varepsilon$, so the true gradient is u . Integrated error on uniform test states is $(\hat{k} - 1)^2/3$. The naturally induced volatility error is 2ε times this error. These identities explain why the experiment does not realize a fixed volatility-target error floor.

H.7.4 Smooth neural weights and certified policies

For each state–time envelope, take its first affine line and its successive nonnegative slope jumps at the rational switch locations. Replace each positive-part hinge by the three cubic-ReLU units in Corollary 3.5 of the main paper. The quadratic skip term is $-\lambda q^2$. Thus every weight, shift, and coefficient is specified by `exact_envelopes.json` and the chosen rational τ . The network is C^2 in the inherited contract; time and regime index separate networks in this finite-horizon model. This is not a $C^{1,2}$ space–time solution of the diffusion problem.

The maximum network size is 27 cubic-ReLU units. For all five smoothing levels, exact rational policy evaluation verifies all 264 primary grid states, for 1,320 state-level checks. The policy does not change at any of these states. This exact finite-grid result is stronger than a small sampled training residual, but it does not assert policy equality at every off-grid initial contract. The uniform analytic policy bound covers those initial states and decreases as follows:

m in $\tau = 2^{-m}$	Uniform critic bound	Uniform policy-loss bound
4	1/80	0.077606934
8	1/1280	0.004850433
12	1/20480	0.000303152
16	1/327680	0.000018947
20	1/5242880	0.000001184

The critic bound follows from the exact triangular smoothing inequality, not from sampling. The policy bound is relative to the unchanged primary contract grid. Adding the contract-discretization budget gives a bound relative to continuous contracts on the same stock set. No post hoc target noise, optimizer selection, or claim of large-neural training is involved.

H.7.5 Diffusion and mesh results

The manufactured stopped diffusion, its full parabolic payoff, and its perturbation sequence are specified in the main paper. The residual bound $2.30e + 1.05e^2$ is analytic. Terminal and lateral defects are identically zero. The actor is analytically greedy and feasible throughout the domain. The additional explicit upwind/central scheme uses a nonnegative transition stencil under the displayed CFL condition, exact boundary data, and the actual refined state and time grids. Its numerical node errors and analytic node bounds are different columns. The latter does not certify an interpolated continuous-state actor. The earlier fixed-state inventory action-cover refinement is not used as evidence of space–time convergence.

H.8 Preservation and interpretation of historical material

H.8.1 An immutable historical snapshot

The entire source tree at reviewed commit

a91b707883b2408de58d9392e4fc546e08a00881

is retained as `archive/pre-r4/`. This is a Git-tree identity, not a reconstructed selection of files. The reviewed root manuscript has blob

2569eee2eb7f80f0a5b31ed07be75d5da9e6a0d3

and remains there unchanged, together with its PDF, bibliography, figures, code, tables, reviews, and release package. That root manuscript and companion were the R4 scientific text; R6 is now active and their complete source is preserved at `archive/pre-r6/`. The archive is provenance, not a second set of current theorem claims.

The main theoretical threads are retained with the following explicit treatment. Frozen-valuation embedding is now exact policy-class restriction and expanded-action equivalence. Exposed affine faces remain a comparator-specific result in EC.1, while entropic overlap is derived rather than excluded. Inventory structure is proved for the external-contract primitive; the penalty-only reversal is preserved as an exact counterexample. Strongly concave inner selection remains a conditional computational lemma, alongside the stronger envelope calculation that needs no such concavity. Physical critical-fractile mismatch remains in EC.2, while the actual optimizer is analyzed by the cash-flow identity. FBSDE/value-gradient relationships, adjoint scope, and vanishing-noise conditioning remain in EC.3. The HJB argument is replaced by a boundary-complete stopped-domain proof and a scheme-specific bound. The constructive neural representation and exact finite-model checks are new positive results.

H.8.2 Historical numerical evidence is not discarded or upgraded

The earlier experiment families, SKU and exact-DP replays, uncertainty tables, actor–critic ablations, residual audits, figures, and manifests remain in the snapshot. They concern their original simulators and objectives. They are neither rerun by the R4 script nor pooled with the new contract experiment. Their successful software checks do not establish hypotheses of a theorem for a different model.

For example, the archived large-neural inventory ladder reports a native-HJB ratio of 0.192341487123, normalized-RMSE ratio 0.979300855398, terminal-loss ratio 0.929461677903, greedy-gap 95th percentile 29.326960415, and held-out increment 0.000199407352943. These numbers are retained as sampled diagnostics. The ladder’s literal label `sampled_inventory_gap_closure_pass` is not treated as a mathematical predicate. It provides neither an interior sup enclosure nor a lateral-boundary defect. The absence of those quantities is not replaced by a zero in any R4 certificate.

The archived finite inventory cover has 122 states and 1,220 Bellman nodes. Its nested action cover refines actions while retaining those state–time nodes. That is a finite-model action refinement, not evidence that a continuous state–time mesh tends to zero. Earlier direct- P /learned- Z ablations retain their architecture, optimizer, data, and target-error interpretation; the new identical-data least-squares calculation is a separate result and reports parity.

H.8.3 Software governance outside the theorem stack

Hash consistency, package replay, release parity, validation guards, and reviewer handoff indexes are valuable reproducibility engineering. They are preserved in the historical tree and indexed in the R4 response and

validation record. They are not restated as propositions proving scientific novelty or correctness. The new source entry point, executed result files, table generator, build commands, and artifact hashes make the present claims inspectable without an internal “referee closure” theorem. A failed or unverified scientific hypothesis cannot be converted to a certificate by changing a manifest label.

The revision therefore preserves the original material while giving each result a specific current role. It replaces incorrect statements with proofs and executable counterchecks rather than suppressing the history or asserting that all historical neural diagnostics have become certified.

H.9 Continuous reconfiguration and boundary-complete computation

The continuous-time counterpart allows a physical state c and an internal or contractual state u to evolve under $\alpha = (a, \theta)$. On a domain K , consider

$$dX_s = b(X_s, \alpha_s) ds + \Sigma_\varepsilon(X_s, \alpha_s) dB_s, \quad X = (c, u), \quad \Sigma_\varepsilon = \begin{bmatrix} \sigma_c & 0 \\ 0 & \sqrt{2\varepsilon}I \end{bmatrix} \quad (\text{H.10})$$

in the block-diagonal special case. A controlled internal ordinary differential equation is obtained at $\varepsilon = 0$. A neural differential parameterization can specify its drift; it does not alter the external reward rule. This retained general diffusion is distinct from the explicitly scaled inventory-contract embedding in the revised main paper.

H.9.1 Value gradients, volatility, and conditioning

For a smooth value function v , define column vectors $P = \nabla v$ and $Z = \Sigma_\varepsilon^\top P$. Along an optimal trajectory, before stopping,

$$dY_s = -(r(X_s, \alpha_s) - \rho Y_s) ds + Z_s^\top dB_s, \quad Y_s = v(s, X_s), \quad (\text{H.11})$$

with the appropriate terminal or exit payoff. When the diffusion is control-independent, the maximizing drift–reward expression is $P^\top b + r$. For a controlled diffusion, the Hessian term must also be optimized. We never infer a feedback law from Z alone when the map $P \mapsto \Sigma^\top P$ has a nontrivial kernel. This is the value-process representation underlying the forward–backward viewpoint (Ma et al., 1994); deep BSDE approximations provide one numerical parameterization (Han et al., 2018).

Lemma H.9.1 (Singular inversion and compatible direct gradients). *In the preference block of (H.10), $Z_u = \sqrt{2\varepsilon}P_u$. If $\widehat{Z}_u = Z_u + \eta_Z$ is inverted, then*

$$\mathbb{E} \left\| \widehat{P}_u - P_u \right\|^2 = \frac{\mathbb{E} \|\eta_Z\|^2}{2\varepsilon}. \quad (\text{H.12})$$

A direct error $\widehat{P}_u = P_u + \eta_P$ has mean-square error $\mathbb{E} \|\eta_P\|^2$ without this inversion factor. At $\varepsilon = 0$, Z_u contains no information about P_u . Taking $P_\phi = \nabla v_\phi$ enforces gradient compatibility, but by itself proves neither an HJB residual bound nor policy optimality.

Proof. Subtract the two volatility identities and divide by $\sqrt{2\varepsilon}$. Squaring and taking expectations gives (H.12). The zero-noise assertion follows directly from the linear map. A scalar critic has a conservative gradient wherever it is differentiable; optimality requires the separate residual and action conditions below. $\square$

Equal-scale additive target errors are an assumption of the conditioning comparison, not a consequence of learning from identical observations. In the recorded least-squares experiment, the direct and volatility parameterizations span precisely the same fitted functions. Their recovered gradients agree to floating-point precision. The observed parity is consistent with (H.12): naturally induced volatility error can shrink with the noise level, so a fixed target-error floor need not apply.

H.9.2 A stopped Dirichlet problem

Fix $D = [0, T] \times \bar{K}$ and the stopping time $\tau = T \wedge \inf\{s \geq t : X_s \notin K\}$. On exit before T , pay $\psi(\tau, X_\tau)$; at T , pay $G(X_T)$. The value is

$$V^*(t, x) = \sup_{\alpha} \mathbb{E}_{t,x} \left[\int_t^\tau e^{-\rho(s-t)} r(X_s, \alpha_s) ds + e^{-\rho(\tau-t)} \Phi(\tau, X_\tau) \right], \quad (\text{H.13})$$

where $\Phi = \psi$ on lateral faces and $\Phi = G$ on the terminal face, with compatible corner data. Assume bounded data, compact controls, well-posed stopped dynamics, admissible arbitrarily accurate Hamiltonian selectors, and the dynamic programming principle and comparison for this Dirichlet problem. Uniform ellipticity and a sufficiently regular bounded domain are one sufficient setting; the theorem below also applies when the stated probabilistic assumptions hold for a degenerate model. Write

$$H^\alpha[w] = r^\alpha + b^\alpha \cdot \nabla w + \frac{1}{2} \text{tr}(\Sigma^\alpha \Sigma^{\alpha^\top} D^2 w), \quad F[w] = w_t + \sup_{\alpha} H^\alpha[w] - \rho w.$$

The problem is $F[V^*] = 0$ in the interior, $V^* = \psi$ on the lateral boundary, and $V^*(T, \cdot) = G$. Reflected and state-constrained problems use different boundary operators and are not implicitly substituted for this one.

Theorem H.9.2 (Absolute value and policy certificates). *Let $w \in C^{1,2}(D)$ satisfy the integrability conditions for stopped Itô's formula. Let A be a measurable admissible actor. Suppose*

$$\begin{aligned} |F[w]| &\leq \delta_H, & 0 &\leq \sup_{\alpha} H^\alpha[w] - H^A[w] \leq \delta_A \quad \text{in the interior,} \\ |w(T, x) - G(x)| &\leq \delta_T, & |w(s, x) - \psi(s, x)| &\leq \delta_\partial \quad \text{on lateral faces.} \end{aligned}$$

Put $\delta_B = \max(\delta_T, \delta_\partial)$ and $H_\rho(s) = (1 - e^{-\rho s})/\rho$ for $\rho > 0$, with $H_0(s) = s$. Then

$$|w(t, x) - V^*(t, x)| \leq \delta_B + H_\rho(T - t)\delta_H, \quad (\text{H.14})$$

$$0 \leq V^*(t, x) - V^A(t, x) \leq 2\delta_B + H_\rho(T - t)(2\delta_H + \delta_A). \quad (\text{H.15})$$

If instead the certified residual is $|w_t + H^A[w] - \rho w| \leq \tilde{\delta}_H$, a direct bound is

$$V^* - V^A \leq 2\delta_B + H_\rho(T - t)(2\tilde{\delta}_H + \delta_A). \quad (\text{H.16})$$

A sequence whose relevant absolute defects vanish therefore has uniformly convergent values and asymptotically optimal actors on this stopped problem.

Proof. For any admissible control, stopped Itô's formula gives

$$J^\alpha - w(t, x) = \mathbb{E} \left[e^{-\rho(\tau-t)} (\Phi - w)(\tau, X_\tau) + \int_t^\tau e^{-\rho(s-t)} (w_t + H^\alpha[w] - \rho w)(s, X_s) ds \right].$$

The integrand is at most δ_H for every control. Approximate measurable maximizers make it at least $-\delta_H$ up to an arbitrarily small error. The boundary contribution has absolute value at most δ_B , proving (H.14). For actor A , the integrand lies between $-\delta_H - \delta_A$ and δ_H , so $w - V^A \leq \delta_B + H_\rho(T - t)(\delta_H + \delta_A)$. Adding this comparison to the upper bound on $V^* - w$ proves (H.15). In particular, both boundary comparisons must be counted.

For (H.16), the actor integrand has absolute value at most $\tilde{\delta}_H$, while every competing integrand is at most $\tilde{\delta}_H + \delta_A$. Applying the same identity once to a competitor and once to A gives the displayed result. Bounded stopping horizons justify the use of $H_\rho(T - t)$. $\square$

A sampled root-mean-square residual is not a bound on δ_H . A deterministic validation cover can supply one only with an independently justified modulus: if $F[w]$ is L_R -Lipschitz and the cover has fill distance h , then $\delta_H \leq \max_{\text{cover}} |F[w]| + L_R h$. Terminal and lateral defects require their own covers and moduli. The action gap needs a separate enclosure over all admissible actions. EC.3 retains these verification details and a counterexample to omitting the lateral boundary.

H.9.3 A constructive instance and a scheme-specific bound

Consider $T = 1$, $K = (-1, 1)^2$, controls $a, \theta \in [-1, 1]$, no discount, and

$$\mathcal{L}^{a, \theta} = a\partial_c + \frac{\theta}{4}\partial_u + 0.045\partial_{cc} + 0.08\partial_{uu}, \quad (\text{H.17})$$

$$r(c, u, a, \theta) = -a^2 - \frac{5}{4}\theta^2 - \frac{1}{4}c^2 - \frac{1}{80}u^2 - 0.125. \quad (\text{H.18})$$

Pay $(c^2 + u^2)/2$ on the entire parabolic boundary. Then $V^* = (c^2 + u^2)/2$: the maximizing controls are $a = c/2$ and $\theta = u/10$, their maximized terms cancel the state costs, and the diffusion contribution is 0.125.

For $e \in [0, 1/16]$, define $w_e = V^* + e(1 - t)(1 - c^2)(1 - u^2)$. Its exact greedy actor is $a_e = (w_e)_c/2$, $\theta_e = (w_e)_u/10$, which remains inside the control box. Direct expansion gives

$$\delta_T = \delta_\partial = \delta_A = 0, \quad \delta_H \leq 2.30e + 1.05e^2. \quad (\text{H.19})$$

Indeed, the time and diffusion terms contribute at most $(1 + 0.09 + 0.16)e$, the linear Hamiltonian perturbations at most $(1 + 0.05)e$, and the squared perturbations at most $(1 + 0.05)e^2$. Thus (H.14)–(H.15) give an analytic

uniform certificate sequence as $e \downarrow 0$. This is a manufactured controlled diffusion, not a trained inventory critic.

We also solve a genuinely refined discrete problem using a locally consistent Markov-chain construction of the type developed by Kushner and Dupuis (2001). With spatial step h , forward/backward differences D^+, D^- , central second differences D_h^2 , and $b^+ = \max(b, 0)$, $b^- = \min(b, 0)$, define

$$L_h^{a,\theta} v = a^+ D_c^+ v + a^- D_c^- v + \frac{1}{4}(\theta^+ D_u^+ v + \theta^- D_u^- v) + 0.045 D_{cc,h}^2 v + 0.08 D_{uu,h}^2 v, \\ T_h v = \max_{a,\theta \in A_k} \{v + \Delta t (L_h^{a,\theta} v + r^{a,\theta})\},$$

where $A_k = \{-1, -1 + 2/k, \dots, 1\}$. Boundary nodes receive the prescribed payoff at every time. If

$$\Delta t (0.25h^{-2} + 1.25h^{-1}) \leq 1, \quad (\text{H.20})$$

all transition weights are nonnegative and sum to one.

Proposition H.9.3 (Quantitative refinement for the displayed scheme). *For (H.17)–(H.18), exact terminal and lateral data, and (H.20), the discrete backward solution v_h satisfies, at every state–time node,*

$$|v_h - V^*| \leq (1 - t) \left(\frac{5}{8}h + \frac{9}{4k^2} \right). \quad (\text{H.21})$$

This is a grid-value bound for this scheme and solution, not a certificate for an arbitrary interpolated neural actor.

Proof. Central differences of V^* are exact. The upwind first derivative contributes an additional $h|a|/2$ and $h|\theta|/8$ to the generator, together at most $5h/8$. Nearest grid controls are within $1/k$ of the continuous maximizers. Completing squares in the two Hamiltonians bounds their joint quantization loss by $(1+5/4)/k^2$. The exact solution is time-independent, so no temporal truncation term remains for this instance. Hence the one-step defect of V^* is at most $\Delta t(5h/8 + 9/(4k^2))$. Nonnegative transition weights make T_h nonexpansive in the sup norm. Backward induction from the exact parabolic data adds one local defect per step and proves (H.21). $\square$

Unlike a fixed state grid with more actions, this construction refines state, time, and actions together. A more general smooth solution would require the corresponding spatial and temporal truncation terms; merely naming monotonicity, stability, and consistency would not establish (H.21).

H.10 Retained R4 study: Computational study

H.10.1 Protocol, computation, and economic comparisons

The recorded protocol fixes $T = 8$, $\beta = 0.97$, $z \in \{0, 1, 2\}$, and $D_z = z + D_0$, where $D_0 \in \{0, 1, 2, 3\}$ has probabilities $(0.1, 0.3, 0.4, 0.2)$. The transition is

$$P = \begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.15 & 0.70 & 0.15 \\ 0.05 & 0.25 & 0.70 \end{pmatrix}.$$

The coefficients are $c = 0.3$, $h = 0.6$, $p = 0.65$, $\kappa = 1.2$, $m = 1.5$, $\lambda = 0.6$, and $r_z = 0.6 + 1.4z$. Stock is $0, 0.5, \dots, 6$, contract is $0, 0.1, \dots, 1$, and the initial state is $(1, 0.5)$. Demand translation makes $\Delta L \leq \Delta z$, while $\Delta r = 1.4\Delta z > 1.2\Delta z$. The transition rows are stochastically ordered. Thus the structural theorem applies to exactly the model computed.

All five methods use unrestricted state-aware stock tables. Joint contract and expanded-action MDP enumerate all 143 action pairs. Frozen contract fixes the economic contract at 0.5 and has 13 distinct stock actions. The capacity-matched gate enumerates 11 dummy labels for each stock choice but leaves the economic contract and next inherited state at 0.5; it still has only 13 distinct economic actions. The physical oracle minimizes physical cost. Under the recorded greatest-pair convention it selects contract one on contract ties; only its physical-cost comparison is invariant to this implementation convention.

Table 10 reports exact discounted expectations. Joint and expanded-action values and policies agree at every nonterminal state, as do frozen and dummy-gate values and policies. The independent Fraction implementation checks 1,320 state–method records; the monotonicity check covers 416 adjacent state pairs. The hull critic agrees at all 264 states. Exact Bellman, terminal, greedy, and accounting defects are zero for the finite model.

Net reward improves from -5.947058 to -5.002311 , a gain of 0.944747 . Physical cost falls by 0.213204 , from 9.587378 to 9.374174 . Nevertheless, the fill rate falls from 0.925926 to 0.882109 . The physical oracle costs 8.506095 , below both contract policies. These results have a direct economic reading: the joint contract changes the cash-flow value of carrying stock, and it economizes on physical cost partly by accepting more lost demand. Improved welfare is not a synonym for higher service.

The full accounting decomposition appears in Table 11. For example, the joint policy earns more premium but also pays more liability, maintenance, and adjustment. Equation EC.7 accounts for all these changes. No historical total-cost field is relabeled as theta-free physical cost.

The simulation uses 64 batches of 4,096 episodes with common exogenous paths across methods and the recorded seed sequence. Two-sided paired Student intervals use 63 degrees of freedom and quantify simulation error only. The paired joint-minus-frozen net-reward difference is 0.943729 with interval $[0.939821, 0.947636]$; the physical-cost difference is -0.212112 with interval $[-0.214933, -0.209291]$. Exact expectations, rather than Monte Carlo significance, establish the finite-model ordering. Batch-level outputs also retain undiscounted service counts.

H.10.2 Contract refinement, conditioning, and diffusion certificates

Table 12 reports the recorded nested contract grids with 10, 20, 40, 80, and 160 intervals. Each row solves the same stock-grid model exactly. Theorem EC.5.5 provides the continuous-contract enclosure, with error bounds decreasing from 0.047158 to 0.00018421. These are analytic error budgets, not thresholds fitted to observed grid differences. A first-order Lipschitz budget is retained in the machine-readable output for comparison.

The conditioning experiment uses 64 batches of 2,048 identical observations for each parameterization. For u uniform on $[-1, 1]$ and standard normal ξ , the observed payoff is $(u + \sqrt{2\varepsilon}\xi)^2/2$. We fit $ku^2/2 + b$ and $\zeta u^2/(2\sqrt{2\varepsilon}) + b$ by unregularized least squares. The resulting shadow prices are ku and $\zeta u/\sqrt{2\varepsilon}$. Table 13 shows matching mean-square errors; recovered slopes agree to floating-point precision in both the original execution and the independent replay. The volatility design becomes less well-conditioned at the smallest noise level, but the exact least-squares fit does not display a shadow-price accuracy advantage for the direct parameterization. No artificial target-noise floor is added to create one.

We additionally compile the smooth critics of Corollary H.11.1 at $\tau = 2^{-m}$ for $m = 4, 8, 12, 16, 20$. They use at most 27 cubic-ReLU units per state–time critic. Exact rational evaluation checks every one of the 264 nonterminal grid states at each level. All five induced policies coincide with the exact grid optimizer at these states, giving zero observed grid-policy loss. The proved uniform policy budget, which also covers initial inherited contracts between grid points, decreases from 0.0776069 to 1.18419×10^{-6} . EC.4 reports the weights, finite-model interpretation, and complete audit locations. This is constructive approximation and validation, not a newly trained neural solver.

Table 14 instantiates (H.19) at the five recorded perturbation levels. It reports separate terminal and lateral defects and the absolute policy-loss bound. The bound decreases to approximately 4.38690×10^{-6} . The state–time–action study in Table 15 is an additional deterministic scheme check, separate from the recorded analytic sequence. It refines 25 to 1,089 spatial nodes, 4 to 94 time steps, and 9 to 65 actions per coordinate. The observed all-node error decreases from 0.0445124 to 0.00742613, within the proved scheme-specific bounds. Floating-point observed errors are diagnostics; the bounds follow from the analytic truncation calculation and nonexpansive propagation.

H.10.3 Evidence retained from earlier versions

The complete earlier manuscript, proofs, figures, numerical outputs, release archive, and reproducibility checks remain available in the historical snapshot. EC.5 identifies their relationship to the present paper. Historical large-neural inventory residual ratios and greedy-gap summaries are not reclassified as uniform certificates: they lack the absolute parabolic-boundary and interior enclosures required by Theorem H.9.2. The finite model, the manufactured diffusion, and those sampled neural diagnostics are different mathematical objects. A successful exact calculation on one does not supply missing hypotheses for another.

H.10.4 Switching thresholds and persistence

Let

$$B_t(z, \theta) = r_z \theta - m \theta^2 + U_z(\theta) + \beta \sum_{z'} P_{zz'} V_{t+1}(z', \theta).$$

This notation removes the current adjustment term, but its continuation still depends on the adjustment-cost parameter. Write $b_i = B_t(z, \theta_i) - \lambda \theta_i^2$ for a finite ordered menu.

Proposition H.10.1 (Thresholds and two-tier hysteresis). *For $\lambda > 0$, adjacent surviving lines $i < j$ of the upper hull exchange optimality at*

$$q_{ij} = \frac{b_i - b_j}{2\lambda(\theta_j - \theta_i)}. \quad (\text{H.22})$$

The greatest optimizer selects the higher tier at equality. Suppose the menu consists of two tiers $a < b$ and put $\Delta = B_t(z, b) - B_t(z, a)$ and $d = b - a$. From inherited tier a the provider switches upward exactly when $\Delta \geq \lambda d^2$. From inherited tier b the provider switches downward exactly when $\Delta < -\lambda d^2$. Thus the interval $-\lambda d^2 \leq \Delta < \lambda d^2$ preserves the inherited tier.

Proof. Subtract the two affine maximands in (EC.17) and solve their equality for q . Ordered slopes and deletion of dominated lines give ordered crossings. With two tiers, the upward value difference at $q = a$ is $\Delta - \lambda d^2$; at $q = b$ it is $\Delta + \lambda d^2$. The strict and weak inequalities follow from selecting the greater tier at a tie. $\square$

The hysteresis width is $2\lambda d^2$ in continuation-adjusted reward units. This is an implementable rule: change tier only when its incremental premium, liability, maintenance, and continuation advantage crosses the appropriate adjustment threshold. It is not a claim that a threshold expressed in an arbitrary economic signal grows monotonically with λ : B_t changes with λ too. For a continuous menu, a quadratic adjustment cost generally gives small continuous adjustments rather than a positive inaction band. For a larger discrete menu, a pairwise comparison is a decision threshold only when both lines survive the hull. These distinctions connect the present structure to, rather than replace, the hysteretic-control literature (Hipp and Holzbaaur, 1988).

Proposition H.10.2 (Aggregate comparative statics). *Fix the transition law, initial state, discount, horizon, and a parameter-independent feasible policy class. If a stage coefficient a enters reward as $-ae_t$, then any optimizers π_1, π_2 at $a_1 < a_2$ satisfy*

$$\mathbb{E}^{\pi_2} \sum_t \beta^t e_t \leq \mathbb{E}^{\pi_1} \sum_t \beta^t e_t.$$

The optimal value is convex and nonincreasing in a when $e_t \geq 0$. Consequently optimal discounted squared movement is nonincreasing in λ , tier square in m , and indemnity exposure $\theta L_z(S)$ in κ . A uniform additive premium increase makes optimal discounted tier usage nondecreasing.

Proof. Write a policy's objective as $A(\pi) - aE(\pi)$. Optimality at the two coefficients gives $A_1 - a_1 E_1 \geq A_2 - a_1 E_2$ and $A_2 - a_2 E_2 \geq A_1 - a_2 E_1$. Adding yields $(a_2 - a_1)(E_1 - E_2) \geq 0$. A supremum of affine

functions is convex; their nonpositive slopes give monotonicity. Replace $-aE$ by $+aE$ for the premium statement. $\square$

These are policy-level exposure results, not pointwise monotonicity of each action. In particular, the κ comparison does not hold automatically when changing κ also changes the participation-feasible set through (3). Neither discounting nor regime persistence is a linear coefficient of a fixed policy's discounted occupation measure; their effects are reported by recomputation, without a universal sign assertion.

A constant contract can be optimal even with state variation. If one tier $\bar{\theta}$ maximizes $r_z\theta - m\theta^2 + U_z(\theta)$ for every regime and $q_0 = \bar{\theta}$, selecting it forever attains every period's intrinsic upper bound with zero adjustment. It is therefore optimal for every horizon, discount, transition law, and $\lambda \geq 0$. Conversely, premium–liability compatibility is substantive. For $T = 1$, $\mathcal{S} = \{0\}$, deterministic $D_z = z$, $r_z = 1$, $q = 1/2$, $m = 3/2$, and $\lambda = 3/5$, the continuous optimizer is

$$\theta^*(z) = \text{proj}_{[0,1]} \frac{1.6 - 1.2z}{4.2}.$$

It decreases from $8/21$ to $2/21$ and then to zero as $z = 0, 1, 2$. This example has a positive external premium; failure is not confined to a self-selected penalty with no counterparty. Compatibility is sufficient, not necessary: a particular calibrated family may remain monotone after the sufficient inequality fails.

H.10.5 Beyond scalar quadratic adjustment

For a continuously differentiable strictly convex potential ϕ on a compact contract interval, define

$$D_\phi(\theta, q) = \phi(\theta) - \phi(q) - \phi'(q)(\theta - q).$$

Replace $\lambda(\theta - q)^2$ by $\lambda D_\phi(\theta, q)$; $\phi(q) = q^2$ recovers the original model. The cost can be asymmetric. This is a specified nonquadratic cost family, not a representation of every possible switching cost.

Theorem H.10.3 (Dual-coordinate envelope). *Suppose current q enters only the adjustment cost and the next inherited state equals θ . Other state transitions and rewards may depend on an operating state x , regime, stock, and θ , but not separately on q . Then*

$$V_t(x, z, q) = \lambda\{\phi(q) - q\phi'(q)\} + \max_{\theta} \{b_t(x, z, \theta) + \lambda\theta\phi'(q)\}, \quad (\text{H.23})$$

where b_t includes the optimization over physical actions and all continuation terms, and subtracts $\lambda\phi(\theta)$. For a scalar ordered finite menu, the same linear-time hull construction and ordered queries apply in the coordinate $y = \phi'(q)$. In dimension d , the identity holds with $q\phi'(q)$ replaced by $q^\top \nabla \phi(q)$ and with an envelope of affine functions of $y = \nabla \phi(q)$.

Proof. Expand $-\lambda D_\phi$ in the Bellman objective. Terms depending solely on q leave the maximum; all remaining dependence on q is $\lambda\theta\phi'(q)$. Convexity makes ϕ' nondecreasing. The scalar affine slopes $\lambda\theta$ are ordered, so the hull proof of Theorem EC.5.3 applies verbatim to ordered dual coordinates. The vector identity follows from the same expansion. $\square$

In dimension greater than one the cells are polyhedral in dual coordinates; the identity does not give a linear-time multidimensional hull algorithm. A continuous physical state changes the intercept computation and may still require numerical integration and physical-state discretization. The theorem accelerates the inherited-contract dimension, not every dimension of a general inventory system.

Nonlinear menu payments need not destroy the structure. If premium and liability are $r_z h(\theta)$ and $\kappa h(\theta) L_z(S)$ for a continuous nondecreasing h , the stock–tier and regime–tier cross differences retain their signs under Assumption EC.5.1; arbitrary continuous tier-only maintenance does not change those cross differences. With quadratic adjustment, Theorem EC.5.2 still applies. The envelope identity also persists because it uses the adjustment term, not linearity of tier payments. However, the particular convexity argument giving the $O(n^{-2})$ bound must be checked anew when h changes. We do not transfer that rate to arbitrary nonlinear payments by assertion.

H.11 Retained constructive smooth neural critics

The exact finite-grid envelope also supplies a smooth neural approximation with a deterministic error budget. This construction uses computed envelope weights rather than statistical training. It provides a certified neural representation of the contract model without making a claim about the optimization of a large network.

Corollary H.11.1 (Smooth value-gradient representation). *For a finite contract grid, let $V_t^{(n)}$ be the exact grid value, extended to every inherited $q \in [0, 1]$ by (EC.17). For each (t, z) write its convex part as*

$$V_t^{(n)}(z, q) + \lambda q^2 = a + bq + \sum_j d_j (q - k_j)^+, \quad d_j \geq 0, \quad \sum_j d_j \leq 2\lambda.$$

For $\tau > 0$, replace each hinge by

$$h_\tau(x) = \frac{(x + \tau)_+^3 - 2x_+^3 + (x - \tau)_+^3}{6\tau^2}$$

and denote the resulting critic by $w_{t,\tau}$; keep $w_{T,\tau} = 0$. It is a quadratic skip connection plus a finite cubic-ReLU network, is C^2 in q , and satisfies

$$0 \leq w_{t,\tau}(z, q) - V_t^{(n)}(z, q) \leq \lambda\tau/3. \quad (\text{H.24})$$

If A_τ exactly maximizes the one-step grid objective with continuation $w_{t+1,\tau}$, then, for $r = T - t$,

$$0 \leq V_t^{(n)}(z, q) - V_t^{A_\tau}(z, q) \leq \beta(\lambda\tau/3)H_{r-1}. \quad (\text{H.25})$$

Combining this bound with (EC.18) bounds loss relative to continuous contract actions. The derivative $\partial_q w_{t,\tau}$ is a compatible shadow price, not an independently fitted vector field.

Proof. The hinge representation follows from the ordered slopes of the upper envelope. Its total slope increase is at most 2λ , the width of the slope interval. The cubic finite difference equals the convolution of

x^+ with the symmetric triangular density $(\tau - |y|)_+/\tau^2$. Convexity gives $h_\tau(x) \geq x^+$, and direct integration shows $h_\tau(x) - x^+ \leq \tau/6$, with equality at $x = 0$. This proves (H.24). Each cubic-ReLU term is C^2 .

The exact and approximate one-step action values differ by a number in $[0, \beta\lambda\tau/3]$. Thus maximizing the latter loses at most $\beta\lambda\tau/3$ in the former, rather than twice that amount, because the approximation is one-sided. The final action uses the exact terminal continuation and has zero loss. Backward policy evaluation gives (H.25). The gradient statement follows from the scalar construction. $\square$

Network size convention. Each hidden unit computes $(wq + b)_+^3$; three units implement one smoothed hinge. The reported maximum of 27 units is per (t, z) critic, excluding its explicit quadratic skip $-\lambda q^2$ and affine skip. There is no sharing across the 24 nonterminal (t, z) critics; their aggregate hidden-unit budget is at most $24 \cdot 27 = 648$, not 27 for one joint network. Each unit has a scalar input weight, bias, and output weight; skip coefficients must also be stored. This convention is not directly comparable to the 1,281 trainable parameters of the joint tanh critic in Section H.4.

H.12 Computational study of accepted adaptivity

We use the same canonical operating primitives throughout the exact comparison: $T = 8$, $\beta = 0.97$, $q_0 = 0.5$, $z_0 = 1$, and $D_z = z + D_0$, where $D_0 = 0, 1, 2, 3$ has probabilities 0.1, 0.3, 0.4, 0.2. The regime transition matrix is

$$P = \begin{pmatrix} .70 & .25 & .05 \\ .15 & .70 & .15 \\ .05 & .25 & .70 \end{pmatrix}.$$

Stock is initially 0, 0.5, $\dots$, 6 and the historical tier menu is 0, 0.1, $\dots$, 1. Coefficients are $c = 0.3$, $h = 0.6$, $p = 0.65$, $\kappa = 1.2$, $m = 1.5$, $\lambda = 0.6$, and $r_z = 0.6 + 1.4z$. Customer service valuation is $v = 2$. A common retainer of 6 can be included on both sides of every provider comparison; the tabulated rewards exclude that constant. The continuous accepted hierarchy permits both stock and tier intervals, using the stock-identification proof rather than an unverified interpolation of the menu optimum.

The initial finite model has 264 nonterminal state nodes and 37,752 stock–tier choices over those nodes. It is small enough for an exact certificate, not presented as a demanding neural benchmark. Its function is to identify economic mechanisms and provide an independently checkable reference. The coupled portfolio experiment has a different function: it tests the accuracy, acceptance, and repeated-solve cost of a learned representation at larger dimension.

H.12.1 Accepted information and menu resolution

Table 1 gives the accepted information decomposition. The time increment is 0.000816876783, the subsequent regime increment is 0.616558016063, and the inherited-tier increment is 0.019290378765. These four optimizations have identical customer, fill, and physical-cost reservations. The continuous full-state optimum has provider reward $-5.233445701435 \dots$ and exactly the static customer’s utility and service. Its exact coefficients are available for direct policy deployment without a neural network.

The older unconstrained information comparison is a different experiment: its inherited-tier increment is 0.050593409253. We retain that result in the companion but do not substitute it for the accepted increment. Likewise, the historical frozen tier of 0.5 is not used as the principal accepted benchmark. The continuous static optimum improves its reward by 0.076946642417 before any adaptivity is introduced.

Table 5 checks whether coarse tiers and randomization generate the accepted benefit. The randomized optimum on 320 intervals is within approximately 1.10×10^{-5} of the exact continuous optimum. A deterministic incumbent on that menu gains 0.63663320 over the continuous static contract. The best continuous protocol needs no lottery at all. The extracted deterministic incumbents are not claimed to be globally optimal on their discrete menus; one less favorable extraction at 20 intervals is retained rather than replaced by an invented monotone sequence.

Table 5: Accepted fine-menu optima and deterministic incumbents

Tier intervals	Randomized optimum	Deterministic incumbent	Incumbent gap
10	-5.244970661	-5.247006823	0.002036162
20	-5.237637537	-5.284466721	0.046829184
40	-5.234637362	-5.236634206	0.001996845
80	-5.233738433	-5.236860106	0.003121673
160	-5.233499396	-5.233703759	0.000204363
320	-5.233456635	-5.233477772	0.000021137

Same customer, fill, and cost reservations as Table 1. Only the first randomized row also has an independent rational R6 certificate. Other menu bounds are floating-point Lagrangian bounds. Deterministic entries are feasible incumbents, not claimed deterministic optima; their nonmonotone gaps reflect extraction, not nonnested feasible sets. The exact continuous optimum is $-5.233445701435 \dots$

H.12.2 Robustness under the actual acceptance constraints

The new sensitivity study solves the accepted full-stock occupation program, not only a provider Bellman relaxation. Each perturbed primitive has a newly optimized continuous static reference. Its fill, customer utility, and physical cost set the reservations unless that row explicitly changes a reservation. When customer utility is raised, we also recompute the accepted static tier before reporting its gain. A comparison with an infeasible old reference would not establish accepted dominance.

The 31-case design varies adjustment cost, maintenance curvature and the maintenance power, the indemnity coefficient, premium intercept and slope, customer service valuation, regime persistence, demand dispersion, demand probabilities, utility and fill reservations, and the stock grid. Tier resolution is varied separately in Table 5. For every feasible LP we record the primal reward, flow residual, constraint slack, reduced-cost residual, and primal–dual gap. These floating-point records are numerical evidence, not rational certificates for every perturbation. Continuous information decompositions are reported only when the stock-support and interior quadratic checks succeed.

Table 6 displays selected cases; EC Table 7 contains every row. Of 31 accepted programs, 29 are feasible; 28 of those exhibit a positive gain over the applicable continuous accepted static comparator. A zero regime-premium slope removes the accepted gain in the reported instance. Increasing the fill floor by

0.015 is infeasible under the original physical-cost cap, and remains infeasible with the tested cap relaxation of 0.6. Both adverse rows are retained. Thus the conclusion is neither universal feasibility nor positive gain for arbitrary service promises.

Table 6: Accepted-problem sensitivity: selected cases

Case	Grid gain	Active	Lottery	Memory gain
canonical	0.625140	FUC	1	0.019290
lam = 0.1	0.728444	FUC	1	0.001115
lam = 4.0	0.361126	FUC	1	0.089863
power = 1.5	0.983858	FUC	1	—
power = 3.0	0.377330	FUC	1	—
persistence = 0.6	0.580915	FUC	1	0.022693
dispersion = 1.2	0.603774	FUC	1	0.018567
utility delta = 0.2	0.585697	FUC	1	0.018221
fill delta = 0.015	infeasible	—	—	—
zero regime premium	0.000000	FUC	0	—

Definitions and all 31 cases appear in EC Table 7. Reservations and static tiers are recomputed within each perturbed problem. The higher-fill case is infeasible under its stated physical-cost cap. The zero-slope case has no accepted gain.

Increasing the maintenance power from two to three leaves a positive accepted grid gain of approximately 0.37733; power 1.5 yields approximately 0.98386. Altering dispersion about the same mean also leaves positive gains in both tested directions. These results complement, rather than prove, Theorem 5.4. The theorem identifies a customer-neutral direction under explicit strict inequalities, whereas the numerical study allows stock changes and solves the full constrained problem. Under stock identification, changing ν alone cancels between the policy and the reservation because filled demand is the same; that invariance is an accounting implication, not new empirical evidence.

The inherited-state increment changes with the commitments and primitives. For example, it is about 0.00111 when $\lambda = 0.1$ and 0.08986 when $\lambda = 4$, even though total accepted grid gain falls in this slice. A cost can reduce the overall benefit of adaptation while making information about the installed tier more valuable. This distinction is obscured by reporting only static-versus-dynamic gains.

H.12.3 Commitment and implementation

The customer-exit program imposes 3,280 continuation constraints on the original eight-review tree. The reported rational protocol satisfies every one, has reward $-5.417032209823\dots$, and retains a gain of $0.453078763224\dots$. Its upper-bound gap is below 6.95×10^{-7} . This result shows directly what survives when customer commitment is relaxed; it does not assume that an ex ante participation check enforces participation later.

For the coupled portfolio study, all 288 submitted decisions are stored as rational tier vectors, together with their nonnegative payment multipliers, graph matrices, customer slack, and loss bounds. An independent checker uses Python integer sparse multiplication, verifies the graph coercivity condition, and recomputes

every fraction without calling an optimizer. At 1,024 services, the maximum learned value–gradient loss bound is 8.0436×10^{-7} per service, compared with a minimum realized gain of approximately 0.016067 per service over the optimized uniform tier. This is a decision-relevant bound on the accepted update, not the earlier finite-chain global residual budget.

The learned fit has a modest average loss advantage over its equal-oracle value-only counterpart, but the paired seed intervals contain zero. Conjugate gradients have substantially better accuracy in this experiment. The learned critic nevertheless supplies a compact compatible representation, accepted decisions, transfer across graph sizes, and a measured deployment tradeoff relative to refactorization. The result is a reproducible method comparison, not evidence that every numerical control problem requires a neural method.

H.13 Operational implications

An accepted tier revision reallocates payments and maintenance resources; it is not merely a provider’s change of performance metric. Three ingredients make its value interpretable. The static alternative must be optimized. Customer service and participation must be enforced in the same comparison. Information classes must be optimized separately under those commitments. With those checks, the canonical gain remains positive, its sources can be separated, and costly inherited-state information has a measurable incremental value.

The exact scalar solution, the interim-participation extension, and the learned portfolio update answer successive implementation questions. They use different computational representations but the same acceptance logic. Complete contractibility remains an assumption, private-information incentive constraints remain distinct, and the portfolio benchmark is a coupled accepted review rather than a solution of an unrestricted high-dimensional multistage problem. The retained continuous-rate and diffusion analyses are labeled as extensions, with their boundary and discretization assumptions intact.

Reproducibility. Separate entry-point scripts implement exact information values, robustness, interim participation, portfolio learning, inaction, and independent verification in `revisions/or-r7-20260921/`. The scripts record rational certificates separately from floating-point sensitivity results. Exact quantities are compared as fractions; tolerances are used only for the explicitly numerical programs. Historical source directories remain unchanged. The response document maps every referee concern to the current result and records the remaining scope boundaries.

H.14 Complete accepted sensitivity and learning diagnostics

H.14.1 Perturbations, reservations, and information restrictions

Table 7 reports the entire 31-case design. Unless specified otherwise, one primitive changes and all others remain canonical. For maintenance power p , cost is $1.5\theta^p$. A persistence value ρ_P replaces the transition by $\rho_P P + (1 - \rho_P)\mathbf{1}(.25, .5, .25)$; both tested matrices remain stochastic. Dispersion rescales the four demand support points about mean 1.7, while the two probability perturbations keep the same support and mean

with probabilities (.05, .35, .45, .15) and (.15, .25, .35, .25). Utility changes are absolute discounted-utility increments. Fill changes are increments in the ratio of discounted filled demand to discounted demand. The stock menu runs from zero to six at its specified step.

The continuous static comparator is reoptimized at every primitive. A changed reservation can make that original static tier infeasible; the JSON contains an explicit feasibility flag. For tighter utility reservations with an identified physical policy, the accepted static quadratic program is solved under the new budget, and that feasible value is used in the displayed gain. No positive gap to an infeasible old baseline is called Pareto improvement.

The full stock-tier LP is used in every sensitivity row. Stock is not removed by inserting the provider-relaxation newsvendor solution. A supporting-line search and endpoint checks determine whether the continuous quadratic information hierarchy is also available. Nonquadratic maintenance, slackened physical commitments, or a boundary optimizer may prevent that interior reduction; the continuous decomposition is then left unreported rather than inferred from the unconstrained problem.

Table 7: Full accepted-problem robustness design

Perturbation	Grid gain	Active	Lottery	Memory gain
canonical	0.625140	FUC	1	0.019290
lam = 0.1	0.728444	FUC	1	0.001115
lam = 1.2	0.549602	FUC	1	0.042054
lam = 4.0	0.361126	FUC	1	0.089863
m = 1.2	0.763126	FUC	1	—
m = 1.8	0.535059	FUC	1	0.012662
power = 1.5	0.983858	FUC	1	—
power = 3.0	0.377330	FUC	1	—
kappa = 0.9	0.633115	FUC	1	0.019727
kappa = 1.5	0.610544	FUC	1	0.018823
slope = 1.1	0.389769	FUC	1	0.012399
slope = 1.7	0.900312	FUC	1	—
r0 = 0.3	0.576151	FUC	1	0.016604
r0 = 0.9	0.657317	FUC	1	0.021211
nu = 1.0	0.625140	FUC	1	0.019290
nu = 3.0	0.625140	FUC	1	0.019290
persistence = 0.6	0.580915	FUC	1	0.022693
persistence = 1.2	0.643989	FUC	1	0.015249
dispersion = 0.8	0.633394	FUC	1	0.019744
dispersion = 1.2	0.603774	FUC	1	0.018567
utility delta = -0.2	0.660211	FUC	1	0.020390
utility delta = 0.2	0.585697	FUC	1	0.018221

Perturbation	Grid gain	Active	Lottery	Memory gain
utility delta = 0.8	0.481061	FUC	1	0.015197
fill delta = -0.015	0.625140	UC	1	—
fill delta = 0.015	infeasible	—	—	—
stock step = 1.0	0.625140	FUC	1	0.019290
stock step = 0.25	0.625140	FUC	1	0.019290
lower variance prob	0.633115	FUC	1	0.019727
higher variance prob	0.610544	FUC	1	0.018823
higher service with cost slack	infeasible	—	—	—
zero regime premium	0.000000	FUC	0	—

Grid gain uses the 0.1 accepted menu and the reoptimized continuous static comparator under the same reservations. For increased customer reservations the accepted static tier is reoptimized again. The full JSON records original-static feasibility separately. Active constraints are filled demand (F), customer utility (U), and physical cost (C), using a numerical tolerance of 10^{-7} . Lottery counts are reachable states randomized by the returned LP policy; a positive count does not prove that every optimizer randomizes. Memory gain is the continuous full-state minus time-regime value, computed only when stock identification and the interior quadratic reduction are verified. Dashes do not assert zero. All robustness diagnostics are floating-point, not interval certificates.

H.14.2 Training samples, derivative labels, and uncertainty

The graph critic uses degree twelve and thirteen fitted readout coefficients. Training seeds are 101 through 110. Each seed generates 96 graphs of size 32 or 64 and three scalar observations at each graph’s central and perturbed inherited-state vectors. Value-only training uses all 288 scalar observations. Value–gradient training uses the identical observations and adds their centered directional-difference residual to the fitting objective. It does not receive neighboring exact values unavailable to the comparator.

For the quadratic oracle and quadratic critic features, centered directional differences equal the exact directional derivative for every nonzero step. Consequently the derivative-weighted objective is also a finite-difference regularizer constructed from the same scalar information. Calling these two descriptions independent baselines would double count one method. The sensitivity file evaluates steps 0.01, 0.05, and 0.1 for the predesignated seed 101 on all four 1,024-service test graphs. The main step is 0.025. There is no teacher spatial grid or discretization-dependent differentiation rule in this experiment; the historical continuous-chain experiment retains that separate limitation.

Test graphs use the fixed seed 9091, four graphs per size, and coupling strengths one through four. Their topology is a cycle with an additional random matching. Tariff numerators are sampled from the integers 36 through 316 and divided by 100. The matrix is supplied in integer CSR form; the test arrays, deployed tier numerators, dual numerators, and fitted coefficients are retained. Training graphs and test graphs are distinct. The model is not retrained for 256 or 1,024 coordinates.

Table 8: Equal-information paired training-seed comparison

d	Mean improvement	Lower 95%	Upper 95%	Favorable seeds
64	3.07e-09	-1.08e-09	7.22e-09	6/10
256	1.76e-09	-1.53e-09	5.04e-09	6/10
1024	1.58e-09	-3.93e-09	7.08e-09	6/10

Improvement is value-only loss minus value–gradient loss per service. The Student intervals summarize variation across ten independent training seeds conditional on the four selected test graphs; they are not population guarantees across arbitrary operations networks. The unit is a training seed, not forty independent replications. Every interval includes zero.

The paired uncertainty calculation averages the four test graphs within a training seed before computing a Student interval across the ten seeds. It does not count all forty deployments as independent training replications. All three intervals contain zero. The result supports a controlled implementation comparison, not a broad claim of statistically established gradient-supervision superiority.

Table 9: All coupled portfolio sizes

d	Method	Mean loss/service	Max. bound/service	Time (ms)
64	Conjugate gradient	7.50e-10	4.88e-09	0.463
64	Analytic Chebyshev	2.57e-08	3.49e-07	0.206
64	Dense Cholesky	0.00e+00	5.61e-12	0.144
64	Sparse direct	0.00e+00	5.61e-12	0.226
64	Learned value–gradient	1.27e-08	3.64e-07	0.186
64	Learned value only	1.58e-08	5.38e-07	0.188
256	Conjugate gradient	8.62e-10	4.92e-09	0.676
256	Analytic Chebyshev	2.75e-08	3.82e-07	0.428
256	Dense Cholesky	0.00e+00	7.25e-12	0.775
256	Sparse direct	0.00e+00	7.25e-12	0.805
256	Learned value–gradient	1.50e-08	5.79e-07	0.409
256	Learned value only	1.68e-08	7.45e-07	0.412
1024	Conjugate gradient	6.11e-10	3.46e-09	1.524
1024	Analytic Chebyshev	2.69e-08	3.67e-07	1.220
1024	Dense Cholesky	0.00e+00	7.80e-12	18.010
1024	Sparse direct	0.00e+00	7.80e-12	7.619
1024	Learned value–gradient	1.52e-08	8.04e-07	1.214
1024	Learned value only	1.68e-08	9.01e-07	1.222

Same definitions and protocol as main Table 1. No favorable-seed selection is used.

H.14.3 Arithmetic and timing scope

Timing uses one CPU environment with BLAS and OpenMP thread counts set to one. Each recorded time is the median of three calls containing the two required linear solves or graph-filter applications and customer-budget calibration. Offline sample generation and fitting, graph construction, and the rational post-deployment audit are excluded from that deployment time. Average value–gradient sample generation

plus training is approximately 0.087 seconds in this environment. Hardware timings are observations, not complexity guarantees.

The learned readout stores thirteen float64 coefficients, or 104 bytes, in addition to the graph and the working vectors needed by every method. A dense matrix alone at 1,024 coordinates stores 8 MiB in float64; this is an accounting comparison, not a claim that a sparse direct solver uses a dense matrix. The graph and its factorization are not free. Sparse direct, dense Cholesky, conjugate gradients with relative tolerance 10^{-4} , and a fixed analytic degree-twelve Chebyshev inverse use the same deployment graph, tariffs, and acceptance calibration.

Each tier and multiplier is rounded to denominator 10^{10} . A downward correction ensures the payment inequality using exact fractions, rather than a floating tolerance. The independent checker confirms matrix symmetry, nonpositive off-diagonal entries, and a row sum of 21 for the integer matrix. Thus the matrix divided by ten is bounded below by 2.17. It recomputes the residual norm and dual slack with arbitrary-precision integer arithmetic. Stored reward and bound fractions must match exactly. The certification concerns the deployed rounded policy and the continuous-action quadratic problem; it neither treats floating learned weights as exact real numbers nor certifies a general continuum HJB.

H.15 Result-specific novelty and evidence map

The stock-support lemma and accepted quadratic reduction establish an economic identification result. Convexity and Riccati algebra are standard; the new assertion is that the same physical commitments force a policy reduction and permit an exact accepted information decomposition. The local perturbation theorem supplies primitive conditions for a customer-neutral improvement. The continuation program and its feasible rational policy quantify the cost of allowing customer exit.

The lattice theorem uses standard monotone-selection machinery under contract-specific compatibility. The ordered envelope exploits a known one-dimensional upper-hull construction; the quadratic grid theorem has its stated contract-discretization scope. Multi-tier inaction intervals use every competing action. The Bregman formula is a dual-coordinate identity, not a fast multidimensional hull theorem. The occupation formulation and Lagrangian upper bound use standard constrained-control duality; their role is independent certification of the economic comparison.

The learned graph critic combines an established polynomial representation with compatible value derivatives and acceptance calibration. Its architecture and least-squares readout are not claimed as new general-purpose learning algorithms. The new evaluation is equal in raw information, transfers across service dimensions, and supplies exact post-deployment loss bounds. The old two-seed chain study, manufactured diffusion checks, and least-squares conditioning parity remain separate evidence classes in the following historical sections.

Table 10: Exact discounted operating outcomes

Method	Net reward	Physical cost	Fill rate	Actions
Joint contract	-5.002311	9.374174	0.882109	143
Expanded-action MDP	-5.002311	9.374174	0.882109	143
Frozen contract	-5.947058	9.587378	0.925926	13
Dummy gate	-5.947058	9.587378	0.925926	13
Physical oracle	-11.972030	8.506095	0.703704	143

All expectations are exact rational calculations; displayed decimals are rounded. Actions count distinct economic choices. The physical oracle minimizes physical cost, not the convention-dependent net reward shown here.

Table 11: Contract cash-flow decomposition

Method	Premium	Liability	Maintenance	Adjustment
Joint contract	9.889658	1.001859	4.308149	0.207786
Expanded-action MDP	9.889658	1.001859	4.308149	0.207786
Frozen contract	7.208555	0.865027	2.703208	0.000000
Dummy gate	7.208555	0.865027	2.703208	0.000000
Physical oracle	14.417109	6.920212	10.812832	0.150000

Net reward equals premium minus physical cost, liability, maintenance, and adjustment. The oracle uses contract one under the recorded tie convention.

Table 12: Exact grid values and continuous-contract enclosures

Intervals	Grid value	Loss bound	Value upper bound
10	-5.00231055	0.04715774	-4.95515281
20	-4.99648554	0.01178944	-4.98469611
40	-4.99331402	0.00294736	-4.99036666
80	-4.99288305	0.00073684	-4.99214621
160	-4.99278573	0.00018421	-4.99260152

Theorem EC.5.5 bounds contract discretization for the unchanged stock set. The lower endpoint is the grid value; the upper endpoint adds the proved bound.

Table 13: Identical-data least-squares conditioning experiment

ε	Direct P MSE	P MSE via Z	Direct condition	Z condition
0.1	0.00131353	0.00131353	6.901	3.461
0.01	0.00010902	0.00010902	6.901	2.980
0.001	0.00001052	0.00001052	6.901	7.653
0.0001	0.00000104	0.00000104	6.901	23.719

Means over 64 batches, each with 2,048 pairs. Integrated shadow-price MSE uses uniform test states on $[-1, 1]$. Condition numbers are spectral design-matrix condition numbers.

Table 14: Absolute certificates for the stopped diffusion

m	δ_H bound	δ_T	δ_∂	δ_A	Policy-loss bound
4	0.147851563	0	0	0	0.295703125
8	0.009000397	0	0	0	0.018000793
12	0.000561586	0	0	0	0.001123172
16	0.000035095	0	0	0	0.000070191
20	0.000002193	0	0	0	0.000004387

$e = 2^{-m}$. The entire parabolic boundary is exact, and the actor is analytically greedy. These are manufactured analytic critics, not fitted neural inventory models.

Table 15: Joint space–time–action refinement

Spatial nodes	Time steps	Actions per axis	Node error	Proved bound
25	4	9	0.04451240	0.34765625
81	10	17	0.02567135	0.16503906
289	29	33	0.01404964	0.08032227
1089	94	65	0.00742613	0.03961182

Observed errors are maximized over every computed state–time node. The bounds are supplied by Proposition H.9.3; they do not certify interpolation or a continuous-state actor.

References

- Abbasi B, Bruzda J, Gavirneni S (2022) Optimal operational service levels in vendor managed inventory contracts—an exact approach. *Operations Research Letters* 50(5):610–617. doi:10.1016/j.orl.2022.08.007.
- Altman E (1999) *Constrained Markov Decision Processes* (Chapman & Hall/CRC, Boca Raton, FL).
- Boyd S, Vandenberghe L (2004) *Convex Optimization* (Cambridge University Press, Cambridge).
- Caggiano KE, Jackson PL, Muckstadt JA, Rappold JA (2007) Optimizing service parts inventory in a multiechelon, multi-item supply chain with time-based customer service-level agreements. *Operations Research* 55(2):303–318. doi:10.1287/opre.1060.0345.
- Czarnecki WM, Osindero S, Jaderberg M, Swirszcz G, Pascanu R (2017) Sobolev training for neural networks. *Advances in Neural Information Processing Systems* 30. arXiv:1706.04859.
- de Farias DP, Van Roy B (2003) The linear programming approach to approximate dynamic programming. *Operations Research* 51(6):850–865. doi:10.1287/opre.51.6.850.24925.
- Defferrard M, Bresson X, Vandergheynst P (2016) Convolutional neural networks on graphs with fast localized spectral filtering. *Advances in Neural Information Processing Systems* 29.
- Federgruen A, Heching A (1999) Combined pricing and inventory control under uncertainty. *Operations Research* 47(3):454–475.
- Han J, Jentzen A, E W (2018) Solving high-dimensional partial differential equations using deep learning. *Proceedings of the National Academy of Sciences* 115(34):8505–8510.
- Henig M, Gerchak Y, Ernst R, Pyke DF (1997) An inventory model embedded in designing a supply contract. *Management Science* 43(2):184–189. doi:10.1287/mnsc.43.2.184.
- Hipp SK, Holzbaur UD (1988) Decision processes with monotone hysteretic policies. *Operations Research* 36(4):585–588. doi:10.1287/opre.36.4.585.
- Hoeffding W (1963) Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association* 58(301):13–30. doi:10.1080/01621459.1963.10500830.
- Hosseinfard Z, Shao L, Talluri S (2022) Service-level agreement with dynamic inventory policy: The effect of the performance review period and the incentive structure. *Decision Sciences* 53:802–826. doi:10.1111/deci.12506.
- Katok E, Thomas D, Davis A (2008) Inventory service-level agreements as coordination mechanisms: The effect of review periods. *Manufacturing & Service Operations Management* 10(4):609–624. doi:10.1287/msom.1070.0188.
- Kushner HJ, Dupuis P (2001) *Numerical Methods for Stochastic Control Problems in Continuous Time*, 2nd ed. (Springer, New York).

- Liang L, Atkins D (2013) Designing service level agreements for inventory management. *Production and Operations Management* 22(5):1103–1117. doi:10.1111/poms.12033.
- Ma J, Protter P, Yong J (1994) Solving forward-backward stochastic differential equations explicitly—a four step scheme. *Probability Theory and Related Fields* 98:339–359.
- Nasser S, Turcic D (2019) Temporary contract adjustment to a retailer with a private demand forecast. *Management Science* 65(1):209–229. doi:10.1287/mnsc.2017.2942.
- Plambeck EL, Zenios SA (2003) Incentive efficient control of a make-to-stock production system. *Operations Research* 51(3):371–386. doi:10.1287/opre.51.3.371.14954.
- Satija H, Amortila P, Pineau J (2020) Constrained Markov decision processes via backward value functions. *Proceedings of Machine Learning Research* 119:8502–8511.
- Sieke MA, Seifert RW, Thonemann UW (2012) Designing service level contracts for supply chain coordination. *Production and Operations Management* 21(4):698–714. doi:10.1111/j.1937-5956.2011.01301.x.
- Spear SE, Srivastava S (1987) On repeated moral hazard with discounting. *Review of Economic Studies* 54(4):599–617. doi:10.2307/2297484.
- Thomas J, Worrall T (1988) Self-enforcing wage contracts. *Review of Economic Studies* 55(4):541–554. doi:10.2307/2297404.
- Tibshirani RJ, Taylor J (2011) The solution path of the generalized lasso. *Annals of Statistics* 39(3):1335–1371. doi:10.1214/11-AOS878.
- Topkis DM (1978) Minimizing a submodular function on a lattice. *Operations Research* 26(2):305–321.
- Virtanen P, Gommers R, Oliphant TE, et al. (2020) SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* 17:261–272. doi:10.1038/s41592-019-0686-2.
- Wachi A, Sui Y (2020) Safe reinforcement learning in constrained Markov decision processes. *Proceedings of Machine Learning Research* 119:9797–9806.